Data Science & Big Data Analytics

Mini Project Report

On

" COVID-19 Data Analysis"

Submitted in the fulfillment of requirements of Savitribai Phule Pune University

For the degree of **Third Year of Computer Engineering**

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By

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CERTIFICATE

This is to Certify that

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Has satisfactorily completed the requirements of Project under Data Science & Big Data Analytics Mini Project For the degree of **Computer Engineering**

On –

COVID-19 Data Analysis

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Abstract

This project focuses on performing exploratory data analytics on the COVID-19 vaccine statewise dataset sourced from Kaggle. The dataset contains vaccination records across all Indian states and union territories. Using Python libraries such as Pandas and Matplotlib, the project analyzes statewise vaccination coverage for the first and second doses, and examines gender-wise distribution of vaccinations. The goal is to derive meaningful insights from the data that can be useful for policymakers and healthcare professionals. The project demonstrates the application of data science techniques such as data description, aggregation, filtering, and visualization on a real-world public health dataset.

Keywords: Data Science, COVID-19, Vaccination Analytics, Pandas, Matplotlib, Data Visualization, Python, Statewise Analysis, Gender-wise Vaccination, Big Data Analytics.

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Title: COVID-19 Vaccine Statewise Data Analytics

Problem Statement:

Use the following covid_vaccine_statewise.csv dataset and perform following analytics on the given dataset -https://www.kaggle.com/sudalairajkumar/covid19-in-india?select=covid_vaccine_statewise.csv

a. Describe the dataset

b. Number of persons state wise vaccinated for first dose in India

c. Number of persons state wise vaccinated for second dose in India

d. Number of Males vaccinated

e. Number of females vaccinated

Software/ Hardware Requirements:

1) Software Requirements

- Operating System: Windows (7/8/10/11)
- Python IDE: Jupyter Notebook / VS Code / PyCharm
- Python Version: 3.x
- Libraries: Pandas, Matplotlib, NumPy

2) Hardware Requirements (For Jupyter Notebook)

- Memory and disk space required per user: 1GB RAM + 1GB of disk + 0.5 CPU core
- Server overhead: 2-4GB or 10% system overhead (whichever is larger), 0.5 CPU cores, 10GB disk space
- Port requirements: Port 8000 plus 5 unique, random ports per notebook
- Processor: Intel Core i3 or higher (recommended)
- Internet connection: Required for downloading the dataset from Kaggle

Proposed Methodology

The project follows a structured data analytics pipeline as described below:

- 1. Data Collection – Download the covid_vaccine_statewise.csv dataset from the Kaggle repository.
- 2. Data Loading – Import the CSV file into a Pandas DataFrame for processing.
- 3. Data Description – Explore the dataset using .head(), .info(), and .describe() methods to understand its structure and statistics.
- 4. Data Aggregation – Compute statewise totals for first dose, second dose, male, and female vaccinations.
- 5. Data Visualization – Plot bar charts using Matplotlib to visualize top states by first dose administration.
- 6. Insight Generation – Derive conclusions from the analysis to understand vaccination coverage patterns.
- 7. Documentation – Record findings and observations in the project report.

The dataset contains multiple records per state across different dates. Hence, groupby() and last() / max() methods are used to extract the latest vaccination data for each state to avoid duplication.

Programming Language Used: Python

Design:

The system is designed to perform descriptive and exploratory data analysis on the COVID-19 vaccine statewise dataset. The design involves a data processing pipeline that ingests raw CSV data, transforms and aggregates it using Pandas operations, and presents visual insights through Matplotlib charts

THEORY:

Exploratory Data Analysis (EDA) is the process of summarizing and visualizing the main characteristics of a dataset before applying advanced modeling. The primary tools used in this project are:

- **Pandas:** A Python library providing high-performance, easy-to-use data structures and data analysis tools. DataFrames allow efficient data manipulation, grouping, and aggregation.
- **Matplotlib:** A comprehensive library for creating static, animated, and interactive visualizations in Python. Bar charts are used to compare statewise vaccination data.
- **Data Aggregation:** The .sum() method is used to compute cumulative vaccine dose counts across states for first dose, second dose, male, and female categories.
- **Sorting and Filtering:** The .sort_values() method ranks states by vaccination count, enabling identification of top-performing states.

The dataset contains information for all Indian states including the date of reporting, total doses administered, first dose count, second dose count, male doses, female doses, and other demographic breakdowns. The design ensures that the code is modular, readable, and produces clear visual outputs.

Code & Output:

```
In [93]: import pandas as pd
         import matplotlib.pyplot as plt

In [74]: # Load dataset
         data = pd.read_csv("covid_vaccine_statewise.csv")

In [75]: # Show dataset
         print(data.head())
```

```
   Updated On  State  Total Doses Administered  Sessions   Sites   \
0  16/01/2021  India                   48276.0    3455.0   2957.0
1  17/01/2021  India                   58604.0    8532.0   4954.0
2  18/01/2021  India                   99449.0   13611.0   6583.0
3  19/01/2021  India                  195525.0   17855.0   7951.0
4  20/01/2021  India                  251280.0   25472.0  10504.0

   First Dose Administered  Second Dose Administered  \
0                  48276.0                       0.0
1                  58604.0                       0.0
2                  99449.0                       0.0
3                 195525.0                       0.0
4                 251280.0                       0.0

   Male (Doses Administered)  Female (Doses Administered)  \
0                        NaN                          NaN
1                        NaN                          NaN
2                        NaN                          NaN
3                        NaN                          NaN
4                        NaN                          NaN

   Transgender (Doses Administered)  ...  18-44 Years (Doses Administered)  \
0                               NaN  ...                               NaN
1                               NaN  ...                               NaN
2                               NaN  ...                               NaN
3                               NaN  ...                               NaN
4                               NaN  ...                               NaN

   45-60 Years (Doses Administered)  60+ Years (Doses Administered)  \
0                               NaN                             NaN
1                               NaN                             NaN
2                               NaN                             NaN
3                               NaN                             NaN
4                               NaN                             NaN

   18-44 Years(Individuals Vaccinated)  45-60 Years(Individuals Vaccinated)  \
0                                  NaN                                  NaN
1                                  NaN                                  NaN
2                                  NaN                                  NaN
3                                  NaN                                  NaN
4                                  NaN                                  NaN

   60+ Years(Individuals Vaccinated)  Male(Individuals Vaccinated)  \
0                                NaN                        23757.0
1                                NaN                        27348.0
2                                NaN                        41361.0
3                                NaN                        81901.0
4                                NaN                        98111.0

   Female(Individuals Vaccinated)  Transgender(Individuals Vaccinated)  \
0                         24517.0                                  2.0
1                         31252.0                                  4.0
2                         58083.0                                  5.0
3                        113613.0                                 11.0
```

```
4                        153145.0                                24.0

   Total Individuals Vaccinated
0                       48276.0
1                       58604.0
2                       99449.0
3                      195525.0
4                      251280.0

[5 rows x 24 columns]
```

In [76]:
```python
# Describe dataset
print("\nDataset Info:")
print(data.info())
```

```
Dataset Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7845 entries, 0 to 7844
Data columns (total 24 columns):
 #   Column                                Non-Null Count  Dtype
---  ------                                --------------  -----
 0   Updated On                            7845 non-null   object
 1   State                                 7845 non-null   object
 2   Total Doses Administered              7621 non-null   float64
 3   Sessions                              7621 non-null   float64
 4    Sites                                7621 non-null   float64
 5   First Dose Administered               7621 non-null   float64
 6   Second Dose Administered              7621 non-null   float64
 7   Male (Doses Administered)             7461 non-null   float64
 8   Female (Doses Administered)           7461 non-null   float64
 9   Transgender (Doses Administered)      7461 non-null   float64
 10   Covaxin (Doses Administered)         7621 non-null   float64
 11  CoviShield (Doses Administered)       7621 non-null   float64
 12  Sputnik V (Doses Administered)        2995 non-null   float64
 13  AEFI                                  5438 non-null   float64
 14  18-44 Years (Doses Administered)      1702 non-null   float64
 15  45-60 Years (Doses Administered)      1702 non-null   float64
 16  60+ Years (Doses Administered)        1702 non-null   float64
 17  18-44 Years(Individuals Vaccinated)   3733 non-null   float64
 18  45-60 Years(Individuals Vaccinated)   3734 non-null   float64
 19  60+ Years(Individuals Vaccinated)     3734 non-null   float64
 20  Male(Individuals Vaccinated)          160 non-null    float64
 21  Female(Individuals Vaccinated)        160 non-null    float64
 22  Transgender(Individuals Vaccinated)   160 non-null    float64
 23  Total Individuals Vaccinated          5919 non-null   float64
dtypes: float64(22), object(2)
memory usage: 1.4+ MB
None
```

In [77]:
```python
print("\nStatistical Summary:")
print(data.describe())
```

```
Statistical Summary:
       Total Doses Administered      Sessions        Sites   \
count              7.621000e+03  7.621000e+03   7621.000000
mean               9.188171e+06  4.792358e+05   2282.872064
std                3.746180e+07  1.911511e+06   7275.973730
min                7.000000e+00  0.000000e+00      0.000000
25%                1.356570e+05  6.004000e+03     69.000000
50%                8.182020e+05  4.547000e+04    597.000000
75%                6.625243e+06  3.428690e+05   1708.000000
max                5.132284e+08  3.501031e+07  73933.000000

       First Dose Administered  Second Dose Administered  \
count             7.621000e+03              7.621000e+03
mean              7.414415e+06              1.773755e+06
std               2.995209e+07              7.570382e+06
min               7.000000e+00              0.000000e+00
25%               1.166320e+05              1.283100e+04
50%               6.614590e+05              1.388180e+05
75%               5.387805e+06              1.166434e+06
max               4.001504e+08              1.130780e+08

       Male (Doses Administered)  Female (Doses Administered)  \
count               7.461000e+03                 7.461000e+03
mean                3.620156e+06                 3.168416e+06
std                 1.737938e+07                 1.515310e+07
min                 0.000000e+00                 2.000000e+00
25%                 5.655500e+04                 5.210700e+04
50%                 3.897850e+05                 3.342380e+05
75%                 2.735777e+06                 2.561513e+06
max                 2.701636e+08                 2.395186e+08

       Transgender (Doses Administered)   Covaxin (Doses Administered)  \
count                       7461.000000                   7.621000e+03
mean                        1162.978019                   1.044669e+06
std                         5931.353995                   4.452259e+06
min                            0.000000                   0.000000e+00
25%                            8.000000                   0.000000e+00
50%                          113.000000                   1.185100e+04
75%                          800.000000                   7.579300e+05
max                        98275.000000                   6.236742e+07

       CoviShield (Doses Administered)  ...  18-44 Years (Doses Administered)
\
count                     7.621000e+03  ...                      1.702000e+03
mean                      8.126553e+06  ...                      8.773958e+06
std                       3.298414e+07  ...                      2.660829e+07
min                       7.000000e+00  ...                      2.662400e+04
25%                       1.331340e+05  ...                      4.344842e+05
50%                       7.567360e+05  ...                      3.095970e+06
75%                       6.007817e+06  ...                      7.366241e+06
max                       4.468251e+08  ...                      2.243304e+08

       45-60 Years (Doses Administered)  60+ Years (Doses Administered)  \
count                      1.702000e+03                    1.702000e+03
```

```
mean                        7.442161e+06                    5.641605e+06
std                         2.225999e+07                    1.681650e+07
min                         1.681500e+04                    9.994000e+03
25%                         2.326275e+05                    1.285605e+05
50%                         2.695938e+06                    1.805696e+06
75%                         6.969726e+06                    5.294763e+06
max                         1.667575e+08                    1.186927e+08

       18-44 Years(Individuals Vaccinated)  \
count                         3.733000e+03
mean                          1.395895e+06
std                           5.501454e+06
min                           1.059000e+03
25%                           5.655400e+04
50%                           2.947270e+05
75%                           9.105160e+05
max                           9.224315e+07

       45-60 Years(Individuals Vaccinated)  60+ Years(Individuals Vaccinated)
\
count                         3.734000e+03                       3.734000e+03
mean                          2.916515e+06                       2.627444e+06
std                           9.567607e+06                       8.192225e+06
min                           1.136000e+03                       5.580000e+02
25%                           9.248225e+04                       5.615975e+04
50%                           8.330395e+05                       7.887425e+05
75%                           2.499280e+06                       2.337874e+06
max                           9.096888e+07                       6.731098e+07

       Male(Individuals Vaccinated)  Female(Individuals Vaccinated)  \
count                  1.600000e+02                    1.600000e+02
mean                   4.461687e+07                    3.951018e+07
std                    3.950749e+07                    3.417684e+07
min                    2.375700e+04                    2.451700e+04
25%                    5.739350e+06                    5.023407e+06
50%                    3.716590e+07                    3.365402e+07
75%                    7.441663e+07                    6.685368e+07
max                    1.349420e+08                    1.156684e+08

       Transgender(Individuals Vaccinated)  Total Individuals Vaccinated
count                           160.000000                   5.919000e+03
mean                          12370.543750                   4.547842e+06
std                           12485.026753                   1.834182e+07
min                               2.000000                   7.000000e+00
25%                            1278.750000                   7.427550e+04
50%                            8007.500000                   4.022880e+05
75%                           19851.000000                   3.501562e+06
max                           46462.000000                   2.506569e+08

[8 rows x 22 columns]
```

[78]:
```
print(data.columns)
```

```
Index(['Updated On', 'State', 'Total Doses Administered', 'Sessions',
       ' Sites ', 'First Dose Administered', 'Second Dose Administered',
       'Male (Doses Administered)', 'Female (Doses Administered)',
       'Transgender (Doses Administered)', ' Covaxin (Doses Administered)',
       'CoviShield (Doses Administered)', 'Sputnik V (Doses Administered)',
       'AEFI', '18-44 Years (Doses Administered)',
       '45-60 Years (Doses Administered)', '60+ Years (Doses Administered)',
       '18-44 Years(Individuals Vaccinated)',
       '45-60 Years(Individuals Vaccinated)',
       '60+ Years(Individuals Vaccinated)', 'Male(Individuals Vaccinated)',
       'Female(Individuals Vaccinated)', 'Transgender(Individuals Vaccinated)',
       'Total Individuals Vaccinated'],
      dtype='object')
```

In [79]:
```
print(data.columns)
```

```
Index(['Updated On', 'State', 'Total Doses Administered', 'Sessions',
       ' Sites ', 'First Dose Administered', 'Second Dose Administered',
       'Male (Doses Administered)', 'Female (Doses Administered)',
       'Transgender (Doses Administered)', ' Covaxin (Doses Administered)',
       'CoviShield (Doses Administered)', 'Sputnik V (Doses Administered)',
       'AEFI', '18-44 Years (Doses Administered)',
       '45-60 Years (Doses Administered)', '60+ Years (Doses Administered)',
       '18-44 Years(Individuals Vaccinated)',
       '45-60 Years(Individuals Vaccinated)',
       '60+ Years(Individuals Vaccinated)', 'Male(Individuals Vaccinated)',
       'Female(Individuals Vaccinated)', 'Transgender(Individuals Vaccinated)',
       'Total Individuals Vaccinated'],
      dtype='object')
```

In [80]:
```
# Total First Dose
first_dose_total = data['First Dose Administered'].sum()
print("\nTotal First Dose:", first_dose_total)
```

```
Total First Dose: 56505259004.0
```

In [81]:
```
# Total Second Dose
second_dose_total = data['Second Dose Administered'].sum()
print("Total Second Dose:", second_dose_total)
```

```
Total Second Dose: 13517788712.0
```

In [82]:
```
# Total Male Vaccinated
male_total = data['Male (Doses Administered)'].sum()
print("Total Males Vaccinated:", male_total)
```

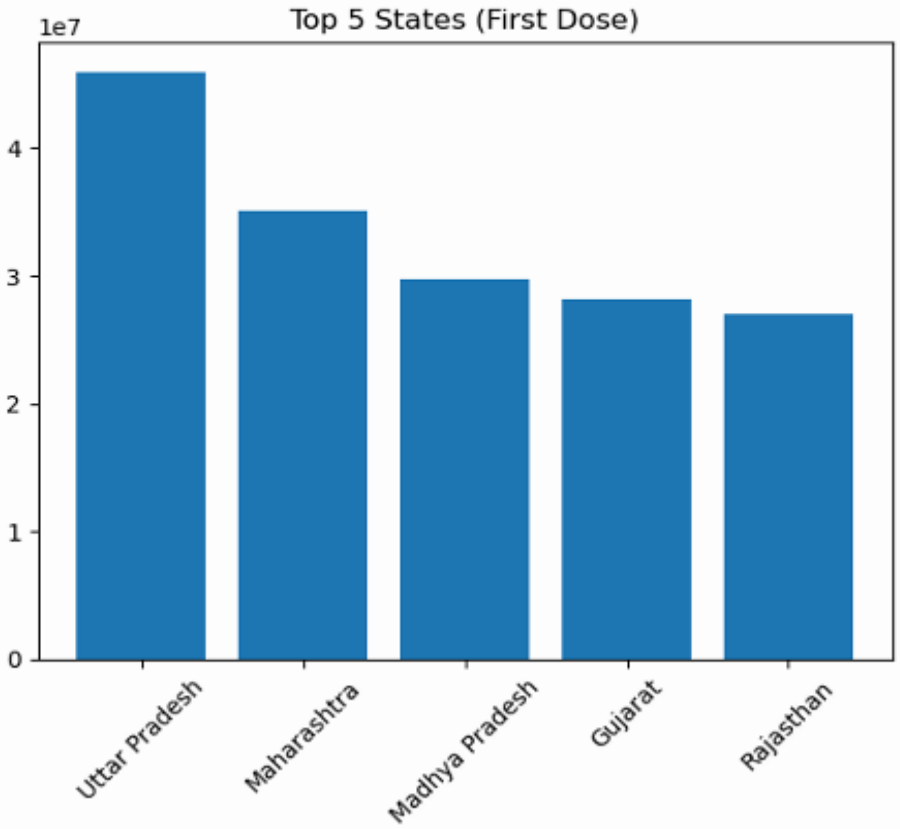
```
Total Males Vaccinated: 27009983996.0
```

In [83]:
```
# Total Female Vaccinated
female_total = data['Female (Doses Administered)'].sum()
print("Total Females Vaccinated:", female_total)
```

```
Total Females Vaccinated: 23639554465.0
```

In [91]:
```
# Plot Top 5 States by First Dose
top_states = data.groupby('State')['First Dose Administered'].max().sort_value
```

In [92]:
```
plt.figure()
plt.bar(top_states.index, top_states.values)
plt.title("Top 5 States (First Dose)")
plt.xticks(rotation=45)
plt.show()
```



Conclusion:

This project demonstrates the use of data science techniques to analyze COVID-19 vaccination data across Indian states using Pandas and Matplotlib. It highlights key insights such as higher vaccination counts in larger states, gaps between first and second doses, and slightly lower female vaccination rates. The visualizations help identify top-performing and underperforming regions, making the analysis useful for improving vaccination strategies and public health decision-making.

References:

- https://www.kaggle.com/sudalairajkumar/covid19-in-india?select=covid_vaccine_statewise.csv
- https://pandas.pydata.org/
- https://matplotlib.org/
- https://scikit-learn.org/stable/
- https://towardsdatascience.com/
- https://www.kaggle.com/
- https://numpy.org/doc/